

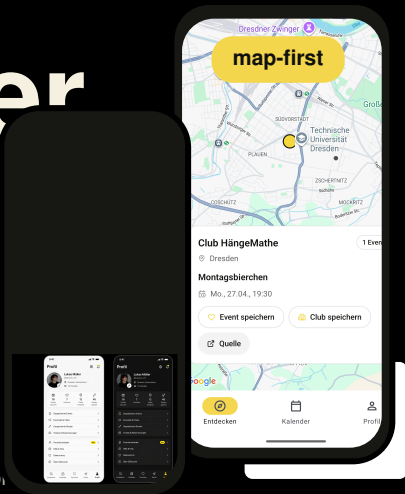
FINAL PROJECT VIDEO

DDiscover

A mobile app for Dresden

HTW Dresden Mobile Networks APL

React Native + Expo Convex Clerk MapLibre



- **CONTEXT**

Student club events need one mobile entry point.

Problem

Event info is split across club pages, feeds, posts, and word of mouth.

Product answer

One map-first mobile flow for place, time, detail, and route.

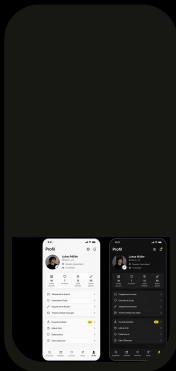
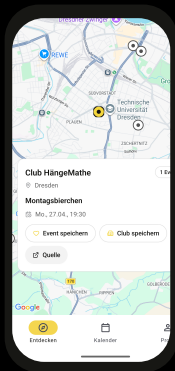
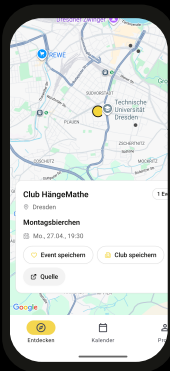
MVP focus

Discover, calendar, details, routes, profile.

Product question: **what is happening nearby, and how do I get there?**

• APP OVERVIEW

The app is organized around the real user flow.



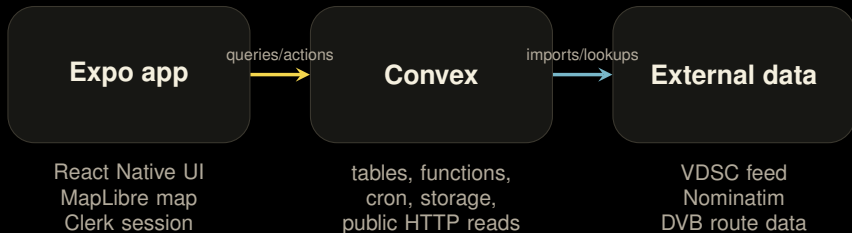
Primary screens

Discover map, calendar, event detail, profile.

- **Discover**
Select a student club on the map
- **Calendar**
Browse upcoming VDSC events
- **Event detail**
Image, source, time, location, route
- **Profile**
Guest-first auth and settings

- **SYSTEM SHAPE**

A small backend owns the hard parts.



Design choice: keep mobile screens thin; let Convex handle imports, normalization, storage, and external service boundaries.

- **IMPORT PIPELINE**

The event data path is idempotent and testable.

VDSC feed
calendar.json

Parser
normalize fields

Upsert
source key

Storage
event images

Cron
every 24h

Structured VDSC feed first. No browser scraping in the core path.

Images are stored in Convex Storage and exposed through generated URLs.

56+ live VDSC events imported during final testing

- LOCATION RELIABILITY

For maps, one wrong coordinate is visible immediately.

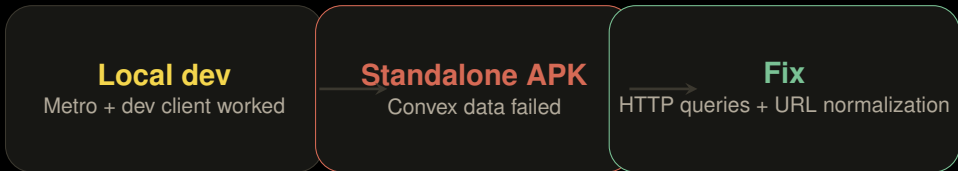
Coordinate priority

- 1 Reviewed Convex**
Manual or checked coordinates win first
- 2 Curated fallback**
Known clubs stay usable offline
- 3 Geocoding cache**
Nominatim only fills gaps

Lesson: automated geocoding helps, but reviewed coordinates are the better source of truth for a student-facing map MVP.

- NATIVE RUNTIME

The hardest bug only appeared in APK builds.



Root causes: unstable Convex function references as effect dependencies, WebSocket path behavior in the APK context, and a trailing slash in the CI Convex URL.

Result: public app data loads through stable Convex HTTP queries; runtime config strips trailing slashes before building API paths.

- TESTING AND CI

Device testing became part of the product work.

17

backend tests

parser, import, geocoding, image

3

core checks

TypeScript, lint, Vitest

2

APK paths

GitHub Actions + GitLab CI

1

phone loop

artifact, install, logs

Expo Go was not enough because the app uses native modules and a dev-client workflow. APK builds make the final behavior testable on a real Android phone and catch missing public configuration before hand-in.

- **DIFFICULTIES**

Most risk came from system boundaries, not screen code.

Android tooling

ADB, emulator disk, stale Metro

Auth handoff

Clerk session vs Convex auth in APK

Data quality

VDSC feed gaps, images, coordinates

Map interaction

Native markers, drawer gestures, theme colors

- **FINAL STATE**

The project has a complete end-to-end discovery loop.

Ready to show

- Map-first student club discovery
- Calendar and event detail pages
- Event images stored through Convex
- DVB route planning on event details
- Guest-first profile and auth shell
- Manual APK builds for phone testing

Closing line: real imported events, mobile UI, backend automation, route planning, and reproducible submission.

What remains

Fully re-enable cloud-synced favorites, add richer filters, verify GitLab hand-in settings, and run final APK smoke test.